D reference: auxiliary supervision, action-only inference



current RGB
legacy scene image

8 EEF states + GT
target pose / motion

noised states + u

Inputs

DINOv3
frozen

scene
meta
flow
action

Shared attention + split FFNs × 4

shared QKV / O
rows query; cols key

	S	M	F	A
S	■			
M	■	■		
F	■		■	
A		■		■

access mask

A reads M + A; no F → A edge.

world/meta
FFN

flow FFN

action FFN

Readouts and execution

Auxiliary objectives (training)

pool 16 meta
deterministic pose → Lpose ⇢ GT pose + masks 50–400 ms

scene-flow
FM field → LFM ⇢ GT flow + masks

action FM field → **Euler ×5**

8 × E7 → execute first → replan

Q1 Replace history encoding

Reference: zero visual-history queries

Q3 Backbone / access / program

Planned D/E + access / generation

Q2 Replace future targets

Solid: data; dashed: supervision